

EXPERIMENT 19

AIM

Detect multiple extensions, executable extensions following document extensions and other deceptive filename patterns without modifying files.

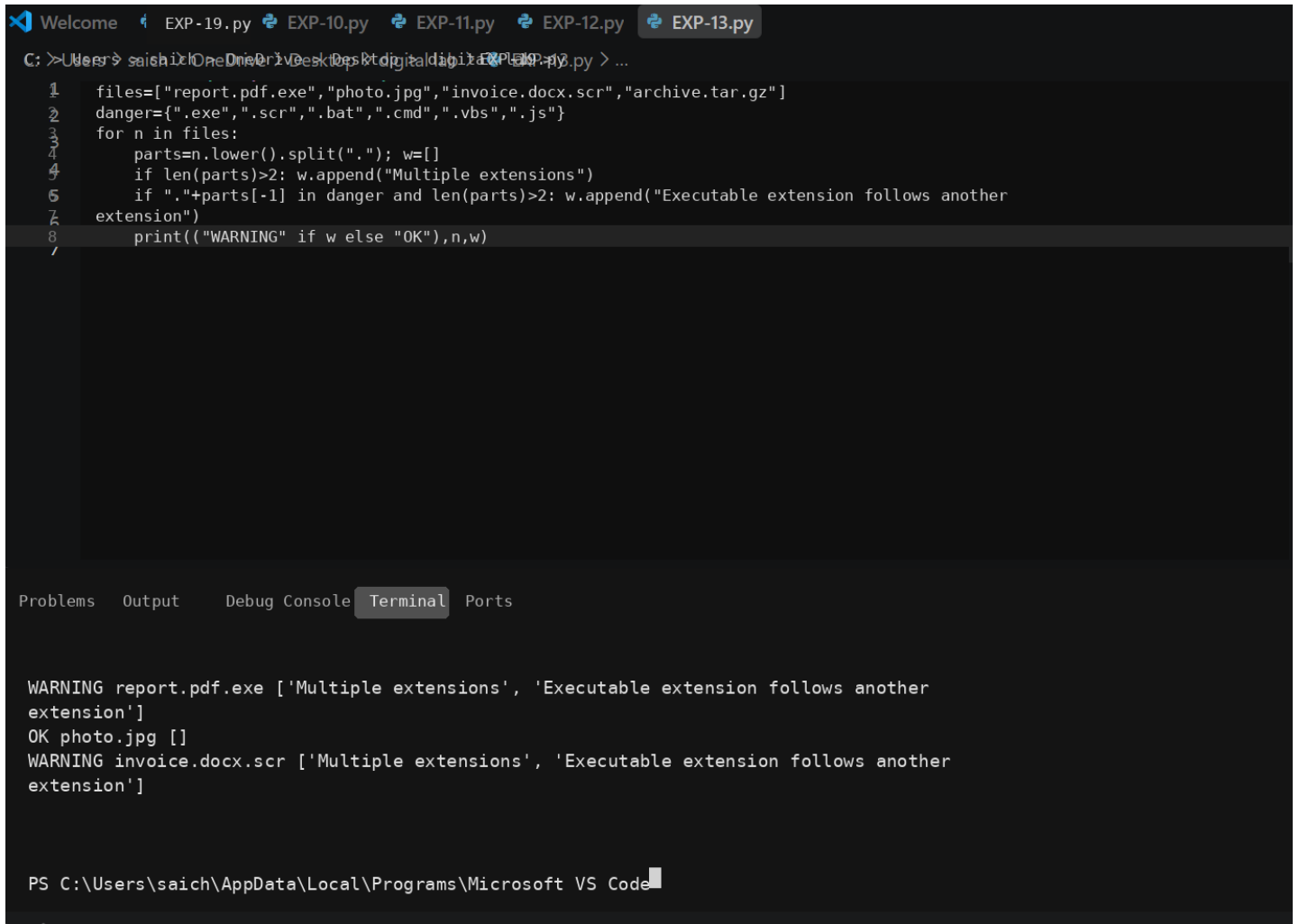
ALGORITHM

1. Store sample filenames and dangerous extensions.
2. Split each filename into components.
3. Check for multiple extensions.
4. Check whether the final extension is executable after another extension.
5. Collect triggered warning rules.
6. Print WARNING with reasons or OK.
7. Stop.

CODE

```
files = [  
    "report.pdf.exe",  
    "photo.jpg",  
    "invoice.docx.scr",  
    "archive.tar.gz"  
]  
  
dangerous = {".exe", ".scr", ".bat", ".cmd", ".vbs", ".js"}  
  
for filename in files:  
    parts = filename.lower().split(".")  
    warnings = []  
  
    if len(parts) > 2:  
        warnings.append("Multiple extensions")  
  
    last_extension = "." + parts[-1]  
    if len(parts) > 2 and last_extension in dangerous:  
        warnings.append("Executable extension follows another extension")  
  
    if warnings:  
        print("WARNING", filename, warnings)  
    else:  
        print("OK", filename)
```

OUTPUT



The screenshot shows a Visual Studio Code editor with a dark theme. At the top, there are tabs for several Python files: 'EXP-19.py', 'EXP-10.py', 'EXP-11.py', 'EXP-12.py', and 'EXP-13.py'. The active file is 'EXP-13.py', which contains the following Python code:

```
1 files=["report.pdf.exe","photo.jpg","invoice.docx.scr","archive.tar.gz"]
2 danger={".exe",".scr",".bat",".cmd",".vbs",".js"}
3 for n in files:
4     parts=n.lower().split("."); w=[]
5     if len(parts)>2: w.append("Multiple extensions")
6     if "."+parts[-1] in danger and len(parts)>2: w.append("Executable extension follows another
7 extension")
8     print(("WARNING" if w else "OK"),n,w)
9
```

Below the editor, the 'Terminal' tab is active, showing the output of the script:

```
WARNING report.pdf.exe ['Multiple extensions', 'Executable extension follows another
extension']
OK photo.jpg []
WARNING invoice.docx.scr ['Multiple extensions', 'Executable extension follows another
extension']
```

The terminal prompt at the bottom indicates the current directory is 'C:\Users\saich\AppData\Local\Programs\Microsoft VS Code'.

RESULT

The program identifies deceptive filename patterns and reports the reason for each warning.